

Test 31: Hatano-Nelson non-Hermitian skin

Verdict: PASS | Generated: 2026-09-02 20:04:01 | Suite: AB-Cloud v19 (Julia 1.12.0)

What this test verifies

Deterministic non-Hermitian spectral property: complex eigenvalue count/pattern.

METHOD: Deterministic non-Hermitian spectral property: complex eigenvalue count/pattern.

Test 31 SERIES: n=5 realizations $\times \{0.5, 0.6, 0.7, 0.8\}$ @48x48, n_cx mean 0,2225,2281,2295, $r \in [0.8953, 0.9253]$, max|Im| 0.3184-0.9576.

Result [SERIES pass 3]: 4 sub-checks, 0 failed ? PASS

- 'plots/plot_01' (.png 600dpi / .svg / .pdf)

Plots

- 'plots/plot_02' (.png 600dpi / .svg / .pdf)

- 'plots/plot_03' (.png 600dpi / .svg / .pdf)

- 'plots/plot_04' (.png 600dpi / .svg / .pdf)

- 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: julia ab_cloud_v19.jl --test 31 --no-two-pass

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

Independent verification guide

[19:58:20.612] [+22215.894s] build_ab_cloud_hamiltonian: 48x48 (N=2304), $\eta=0.5000$, t=1.00, W=1.00, torus, 4 vortices, model=:monumental

[19:58:24.636] [+22219.918s] calc: $r_{NN} = 0.894102$ (n=2304, Poisson-box ref NaN, 10 clouds)

[19:58:48.541] [+22249.824s] calc: $r_{NN} = 0.907204$ (n=2304, Poisson-box ref 0.851687, 10 clouds)

[19:59:02.308] [+22257.590s] calc: $r_{NN} = 0.925299$ (n=2304, Poisson-box ref 0.852026, 10 clouds)

[19:59:20.809] [+22276.091s] calc: $r_{NN} = 0.930754$ (n=2304, Poisson-box ref 0.850043, 10 clouds)

[19:59:21.815] [+22277.097s] build_ab_cloud_hamiltonian: 48x48 (N=2304), $\eta=0.5000$, t=1.00, W=1.00, torus, 4 vortices, model=:monumental

[19:59:25.833] [+22281.115s] calc: $r_{NN} = 0.887760$ (n=2304, Poisson-box ref NaN, 10 clouds)

[19:59:43.930] [+22299.212s] calc: $r_{NN} = 0.910544$ (n=2304, Poisson-box ref 0.852173, 10 clouds)

[20:00:02.298] [+22317.580s] calc: $r_{NN} = 0.923175$ (n=2304, Poisson-box ref 0.852783, 10 clouds)

[20:00:20.812] [+22336.094s] calc: $r_{NN} = 0.930952$ (n=2304, Poisson-box ref 0.850774, 10 clouds)

[20:00:21.952] [+22337.234s] build_ab_cloud_hamiltonian: 48x48 (N=2304), $\eta=0.5000$, t=1.00, W=1.00, torus, 4 vortices, model=:monumental

[20:00:26.458] [+22341.741s] calc: $r_{NN} = 0.898821$ (n=2304, Poisson-box ref NaN, 10 clouds)

[20:00:45.029] [+22360.311s] calc: $r_{NN} = 0.878783$ (n=2304, Poisson-box ref 0.851955, 10 clouds)

[20:01:05.099] [+22380.381s] calc: $r_{NN} = 0.888162$ (n=2304, Poisson-box ref 0.852427, 10 clouds)

[20:01:26.846] [+22402.128s] calc: $r_{NN} = 0.891964$ (n=2304, Poisson-box ref 0.850764, 10 clouds)

[20:01:28.031] [+22403.313s] build_ab_cloud_hamiltonian: 48x48 (N=2304), $\eta=0.5000$, t=1.00, W=1.00, torus, 4 vortices, model=:monumental

[20:01:32.497] [+22407.779s] calc: ?r?_NN = 0.897703 (n=2304, Poisson-box ref NaN, 10 clouds)

[20:01:51.151] [+22426.433s] calc: ?r?_NN = 0.903503 (n=2304, Poisson-box ref 0.851913, 10 clouds)

[20:02:14.218] [+22449.500s] calc: ?r?_NN = 0.923336 (n=2304, Poisson-box ref 0.852584, 10 clouds)

[20:02:39.742] [+22475.024s] calc: ?r?_NN = 0.935506 (n=2304, Poisson-box ref 0.850682, 10 clouds)

[20:02:42.557] [+22477.839s] build_ab_cloud_hamiltonian: 48x48 (N=2304), ?=0.5000, t=1.00, W=1.00, torus, 4 vortices, model=:monumental

[20:02:48.011] [+22483.293s] calc: ?r?_NN = 0.898157 (n=2304, Poisson-box ref NaN, 10 clouds)

[20:03:11.146] [+22506.428s] calc: ?r?_NN = 0.906060 (n=2304, Poisson-box ref 0.851688, 10 clouds)

[20:03:35.069] [+22530.351s] calc: ?r?_NN = 0.920104 (n=2304, Poisson-box ref 0.851832, 10 clouds)

[20:03:59.821] [+22555.103s] calc: ?r?_NN = 0.937177 (n=2304, Poisson-box ref 0.850592, 10 clouds)

CONSOLE CAPTURE

??

TEST 31 (pass 3, SERIES): Hatano?Nelson ? realization ensemble

??

Ensemble: n=5 realizations x? ? {0.5, 0.6, 0.7, 0.8}, ? = 5.0

Lattice: 48x48 (N=2304), W_eff = 1.00 (standard), Nv=4 neutral, q=-1.00

? RUNTIME: 15 zgeev + 5 heevr rows at 48? ? 1?2 min

real 1/5 ?=0.5: n_complex = 0, max|Im| = 0.0000, ?r?NN = 0.8941 (3.3 s)

real 1/5 ?=0.6: n_complex = 2243, max|Im| = 0.3195, ?r?NN = 0.9072 (18.6 s)

real 1/5 ?=0.7: n_complex = 2283, max|Im| = 0.6397, ?r?NN = 0.9253 (18.8 s)

real 1/5 ?=0.8: n_complex = 2298, max|Im| = 0.9602, ?r?NN = 0.9308 (18.5 s)

real 2/5 ?=0.5: n_complex = 0, max|Im| = 0.0000, ?r?NN = 0.8878 (3.3 s)

real 2/5 ?=0.6: n_complex = 2237, max|Im| = 0.3194, ?r?NN = 0.9105 (18.1 s)

real 2/5 ?=0.7: n_complex = 2297, max|Im| = 0.6396, ?r?NN = 0.9232 (18.4 s)

real 2/5 ?=0.8: n_complex = 2299, max|Im| = 0.9598, ?r?NN = 0.9310 (18.5 s)

real 3/5 ?=0.5: n_complex = 0, max|Im| = 0.0000, ?r?NN = 0.8988 (3.8 s)

real 3/5 ?=0.6: n_complex = 2147, max|Im| = 0.3147, ?r?NN = 0.8788 (18.6 s)

real 3/5 ?=0.7: n_complex = 2239, max|Im| = 0.6319, ?r?NN = 0.8882 (20.1 s)

real 3/5 ?=0.8: n_complex = 2274, max|Im| = 0.9501, ?r?NN = 0.8920 (21.7 s)

real 4/5 ?=0.5: n_complex = 0, max|Im| = 0.0000, ?r?NN = 0.8977 (3.7 s)

real 4/5 ?=0.6: n_complex = 2260, max|Im| = 0.3199, ?r?NN = 0.9035 (18.7 s)

real 4/5 ?=0.7: n_complex = 2300, max|Im| = 0.6399, ?r?NN = 0.9233 (23.1 s)

real 4/5 ?=0.8: n_complex = 2303, max|Im| = 0.9601, ?r?NN = 0.9355 (25.5 s)

real 5/5 ?=0.5: n_complex = 0, max|Im| = 0.0000, ?r?NN = 0.8982 (4.6 s)

real 5/5 ?=0.6: n_complex = 2237, max|Im| = 0.3183, ?r?NN = 0.9061 (23.1 s)

real 5/5 ?=0.7: n_complex = 2287, max|Im| = 0.6381, ?r?NN = 0.9201 (23.9 s)

real 5/5 ?=0.8: n_complex = 2303, max|Im| = 0.9578, ?r?NN = 0.9372 (24.7 s)

Ensemble means: ?=0.5: n_cx=0.0, max|Im|=0.00e+00, ?r?=0.8953

?=0.8: n_cx=2295.4, max|Im|=0.9576, ?r?=0.9253 | Poisson-box baseline ? 0.8500

S1 anchor: OK ? | S2 sweep: OK ? | S3 ?r? rise: OK ? (0.8953?0.9253) | S4 max|Im| grow: OK ? (0.3184?0.9576)

??

?=0.5 Hermitian ?real PASS ? n_complex = 0 and max|Im| < 1e-10 in all 5 realizations

n_complex > 0 ?(? ,k) PASS ? every non-Hermitian row (5 ×3) develops complex eigenvalues

?r? rises with ? (2-D repulsion) PASS ? ensemble mean 0.8953 (?=0.5, line) ? 0.9253 (?=0.8, area)

max|Im| grows with ? PASS ? ensemble mean 0.3184 (?=0.6) ? 0.9576 (?=0.8), g = (?%?)•ln|?

Test 31c [SERIES pass 3]: 4 sub-checks, 0 failed ? PASS